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NETWORK



CLOUD IDS IMPLEMENTATION

Intrusion detection system

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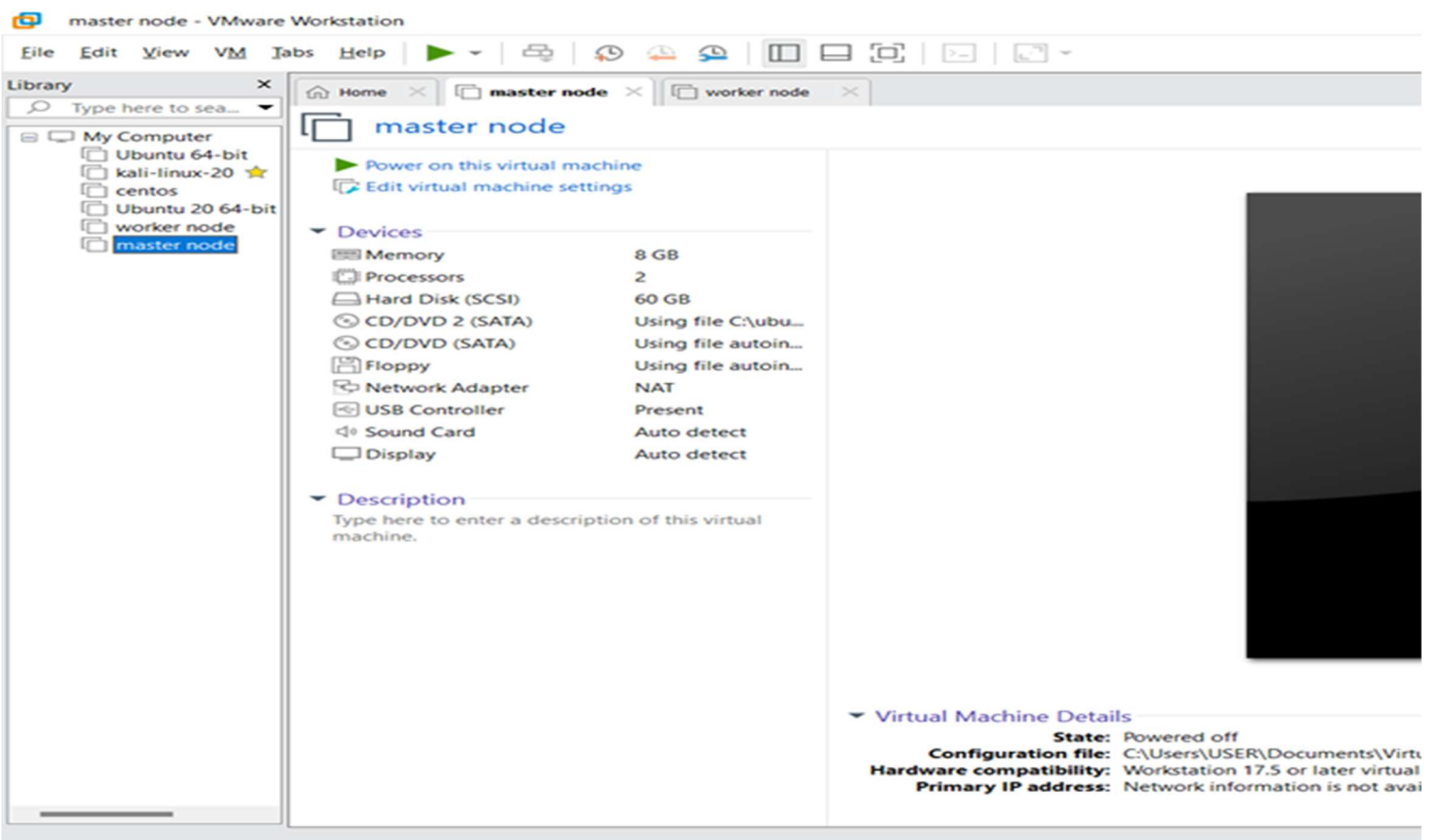
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IMPLEMENTATION :

- we present the practical implementation of our proposed system process consists of:
 1. Two VMs have been created, and a 64-bit Ubuntu operating system image has been downloaded on each VM.
 2. Container images for K3s have been downloaded on both virtual machines, and the master node is considered the server node, while the worker node is the client node.



IMPLEMENTATION :

3. After installing k3s on the server, it was assigned an IP and the NAT scope for the network interface was configured.

K3s status for server and his IP :

```
● k3s.service - Lightweight Kubernetes
   Loaded: loaded (/etc/systemd/system/k3s.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-04-24 14:10:02 CEST; 1 day 8h ago
     Docs: https://k3s.io
  Main PID: 1403 (k3s-server)
    Tasks: 113
   Memory: 783.9M (peak: 1.0G swap: 5.5M swap peak: 5.5M)
      CPU: 58min 25.086s
   CGroup: /system.slice/k3s.service
           └─1403 "/usr/local/bin/k3s server"
              └─1850 "containerd "
                 └─2825 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0>
                    └─2844 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0>
                       └─2976 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0>
                          └─3018 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0>
                             └─3065 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0>
```

Cancel

Wired

Apply

Details

Identity

IPv4

IPv6

Security

Link speed

1000 Mb/s

IPv4 Address

192.168.18.129

IPv6 Address

fe80::20c:29ff:fe0b:97f3

Hardware Address

00:0C:29:0B:97:F3

Default Route

192.168.18.2

DNS

192.168.18.2

☒ Connect automatically

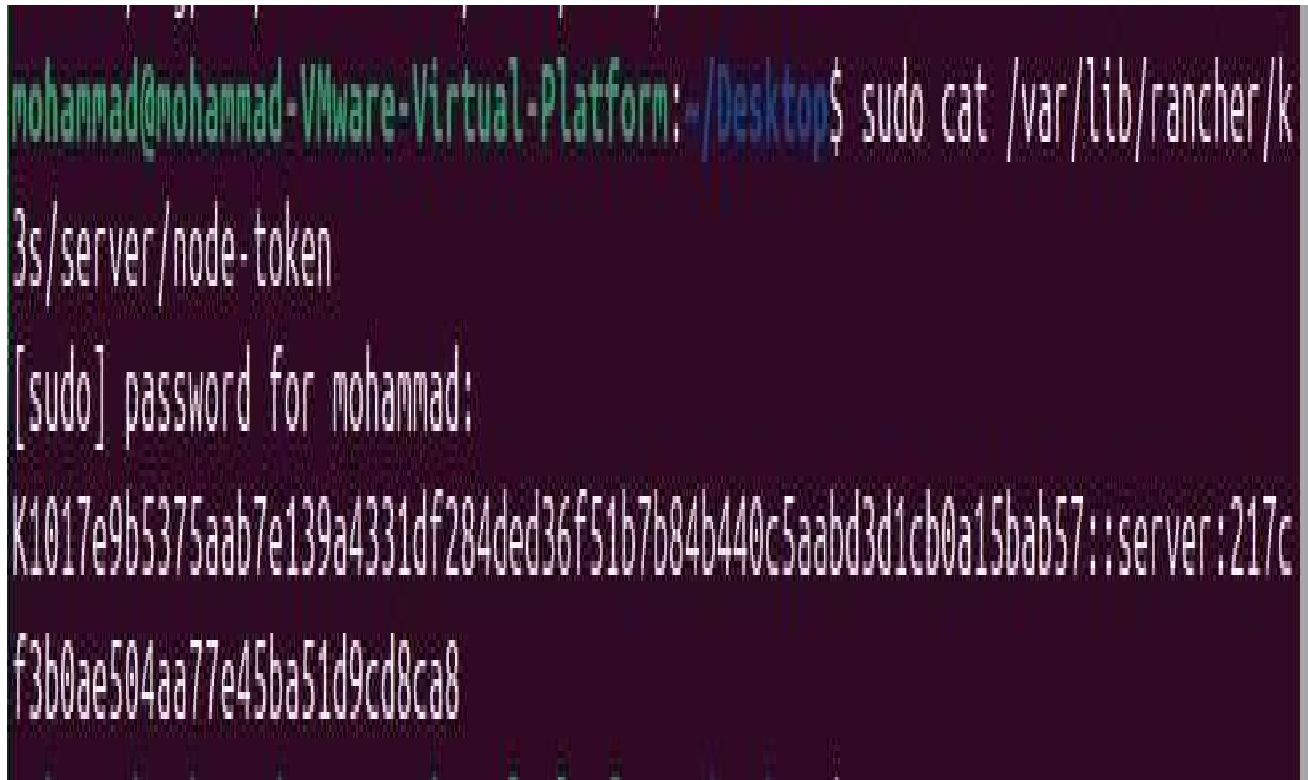
☒ Make available to other users

☐ Metered connection: has data limits or can incur charges
Software updates and other large downloads will not be started automatically.

Remove Connection Profile...

4. After installing k3s on the server and giving it an IP, we want to obtain the token from the server to create a client for it.

Discover the token from server :



```
mohammad@mohammad-Virtual-Platform: ~/Desktop$ sudo cat /var/lib/rancher/k3s/server/node-token
[sudo] password for mohammad:
K1017e9b5375aab7e139a4331df284ded36f51b7b84b440c5aabd3d1cb0a15bab57::server:217cf3b0ae504aa77e45ba51d9cd8ca8
```

Download the client for server with the server's IP and his token :

```
curl -sfL https://get.k3s.io |
K3S_URL=https://192.168.18.129:6443
K3S_TOKEN=K1017e9b5375aab7e139a4331df284ded36f5
1b7b84b440c5aabd3d1cb0a15bab57::server:217cf3b0ae5
04aa77e45ba51d9cd8ca8 sh -
```

K3s status for client and his IP :

```
● k3s-agent.service - Lightweight Kubernetes
   Loaded: loaded (/etc/systemd/system/k3s-agent.service; enabled; preset: enabl>
   Active: active (running) since Fri 2026-04-24 14:46:20 CEST; 8s ago
     Docs: https://k3s.io
   Process: 4612 ExecStartPre=/sbin/modprobe br_netfilter (code=exited, status=0/>
   Process: 4615 ExecStartPre=/sbin/modprobe overlay (code=exited, status=0/SUCCE>
   Main PID: 4617 (k3s-agent)
    Tasks: 33
   Memory: 67.3M (peak: 72.2M)
      CPU: 4.782s
   CGroup: /system.slice/k3s-agent.service
           └─4617 "/usr/local/bin/k3s agent"
             └─4634 "containerd "
               └─5034 /var/lib/rancher/k3s/data/d76b59b2203578bb4cb3438338ccad2f0d3e>
```

Cancel

Wired

Apply

Details

Identity

IPv4

IPv6

Security

Link speed

1000 Mb/s

IPv4 Address

192.168.18.131

IPv6 Address

fe80::20c:29ff:fed6:4582

Hardware Address

00:0C:29:D6:45:82

Default Route

192.168.18.2

DNS

192.168.18.2

☒ Connect automatically

☒ Make available to other users

☐ Metered connection: has data limits or can incur charges

Software updates and other large downloads will not be started automatically.

Remove Connection Profile...

5. Ping from client to server :

```
mohammad@worker1:~/Desktop$ ping 192.168.18.129
PING 192.168.18.129 (192.168.18.129) 56(84) bytes of data.
64 bytes from 192.168.18.129: icmp_seq=1 ttl=64 time=0.656 ms
64 bytes from 192.168.18.129: icmp_seq=2 ttl=64 time=0.597 ms
64 bytes from 192.168.18.129: icmp_seq=3 ttl=64 time=1.41 ms
64 bytes from 192.168.18.129: icmp_seq=4 ttl=64 time=1.17 ms
64 bytes from 192.168.18.129: icmp_seq=5 ttl=64 time=1.53 ms
64 bytes from 192.168.18.129: icmp_seq=6 ttl=64 time=1.51 ms
64 bytes from 192.168.18.129: icmp_seq=7 ttl=64 time=2.15 ms
64 bytes from 192.168.18.129: icmp_seq=8 ttl=64 time=1.86 ms
64 bytes from 192.168.18.129: icmp_seq=9 ttl=64 time=2.22 ms
^C
--- 192.168.18.129 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8074ms
rtt min/avg/max/mdev = 0.597/1.454/2.221/0.547 ms
mohammad@worker1:~/Desktop$
```

6. Ping from server to client :

```
mohammad@mohammad-VMware-Virtual-Platform:~/Desktop$ ping 192.168.18.131
PING 192.168.18.131 (192.168.18.131) 56(84) bytes of data.
64 bytes from 192.168.18.131: icmp_seq=1 ttl=64 time=0.946 ms
64 bytes from 192.168.18.131: icmp_seq=2 ttl=64 time=0.986 ms
64 bytes from 192.168.18.131: icmp_seq=3 ttl=64 time=3.11 ms
64 bytes from 192.168.18.131: icmp_seq=4 ttl=64 time=2.02 ms
64 bytes from 192.168.18.131: icmp_seq=5 ttl=64 time=2.16 ms
64 bytes from 192.168.18.131: icmp_seq=6 ttl=64 time=1.56 ms
64 bytes from 192.168.18.131: icmp_seq=7 ttl=64 time=2.68 ms
64 bytes from 192.168.18.131: icmp_seq=8 ttl=64 time=2.04 ms
64 bytes from 192.168.18.131: icmp_seq=9 ttl=64 time=2.48 ms
64 bytes from 192.168.18.131: icmp_seq=10 ttl=64 time=1.80 ms
64 bytes from 192.168.18.131: icmp_seq=11 ttl=64 time=3.42 ms

--- 192.168.18.131 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10017ms
rtt min/avg/max/mdev = 0.946/2.107/3.415/0.750 ms
^Cmohammad@mohammad-VMware-Virtual-Platform:~/Desktop$
```

7. Falco intrusion detection system, along with its rule file and configuration, has been downloaded on the cloud server or on the master node.

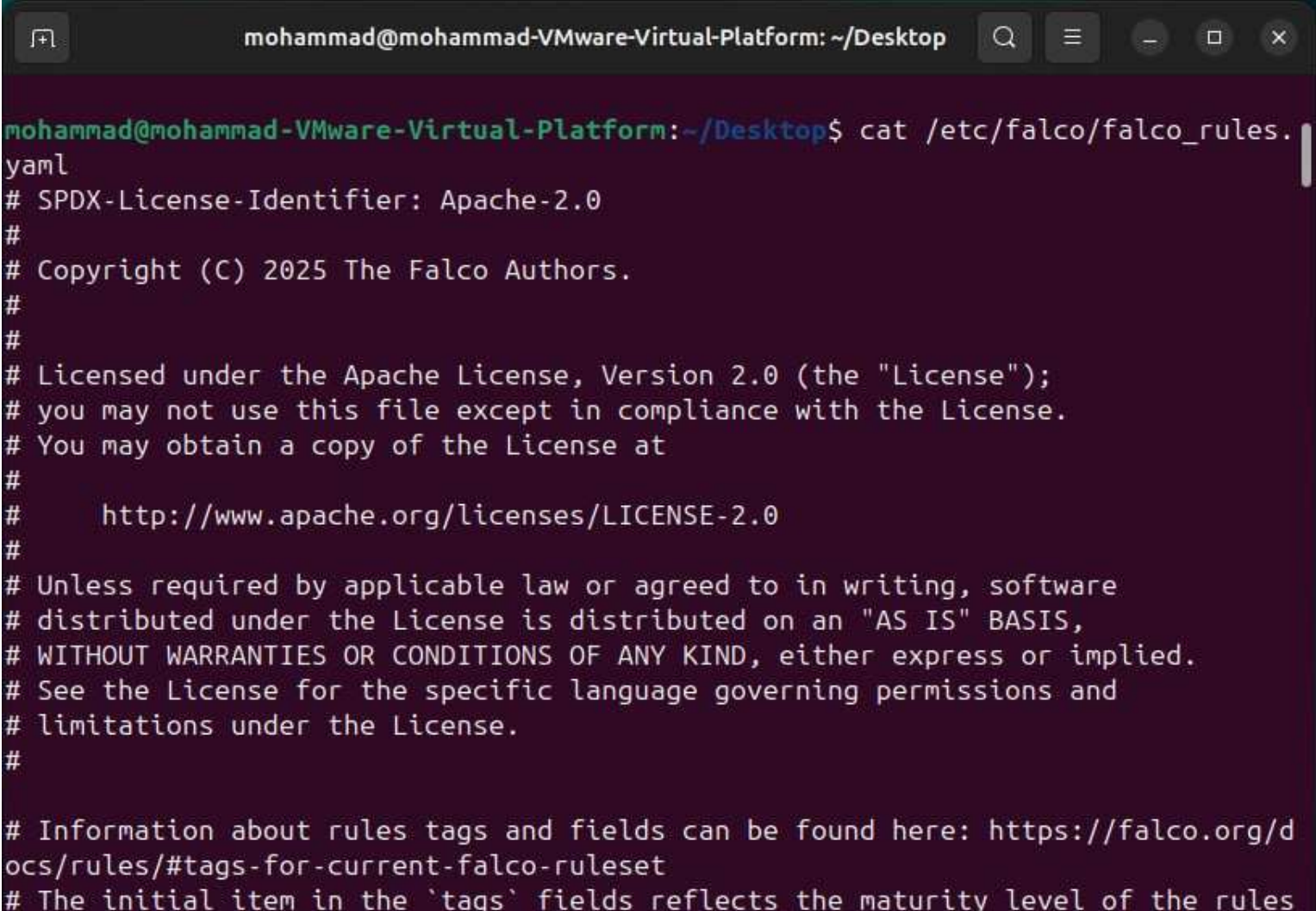
We downloaded from :

<https://falco.org/docs/getting-started/falco-linux-quickstart/>

FALCO IDS STATUS :

```
mohammad@mohammad-VMware-Virtual-Platform:~/Desktop$ sudo systemctl status falco-modern-bpf.service
[sudo] password for mohammad:
● falco-modern-bpf.service - Falco: Container Native Runtime Security with mode>
   Loaded: loaded (/usr/lib/systemd/system/falco-modern-bpf.service; enabled;>
   Active: active (running) since Fri 2026-04-24 15:09:00 CEST; 10min ago
     Docs: https://falco.org/docs/
  Main PID: 8955 (falco)
    Tasks: 15 (limit: 4541)
   Memory: 64.1M (peak: 82.2M)
      CPU: 13.015s
   CGroup: /system.slice/falco-modern-bpf.service
           └─8955 /usr/bin/falco -o engine.kind=modern_ebpf
```


FALCO RULES :

A terminal window with a dark background. The title bar shows 'mohammad@mohammad-VMware-Virtual-Platform: ~/Desktop'. The prompt is 'mohammad@mohammad-VMware-Virtual-Platform: ~/Desktop\$'. The command 'cat /etc/falco/falco_rules.yaml' has been executed. The output shows the license and header information for the Falco ruleset.

```
mohammad@mohammad-VMware-Virtual-Platform: ~/Desktop$ cat /etc/falco/falco_rules.yaml
# SPDX-License-Identifier: Apache-2.0
#
# Copyright (C) 2025 The Falco Authors.
#
#
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#
# Information about rules tags and fields can be found here: https://falco.org/docs/rules/#tags-for-current-falco-ruleset
# The initial item in the 'tags' fields reflects the maturity level of the rules
```

Example for falco rules :

Access sensitive files like shadow that stores user password hashes and readable only by root .

Shadow file : `/etc/shadow`

From the client we can try this instruction:

`Sudo cat /etc/shadow`


```
mohammad@mohammad-VMware-Virtual-Platform: ~/Desktop
Apr 25 22:11:58 mohammad-VMware-Virtual-Platform falco[28756]: [libs]: Trying to
open the right engine!
Apr 25 22:14:23 mohammad-VMware-Virtual-Platform falco[28756]: 22:14:23.49131808
1: Warning Sensitive file opened for reading by non-trusted program | file=/etc/
shadow gparent=sudo gparent=bash gggparent=gnome-terminal- evt_type=openat user
=root user_uid=0 user_loginuid=1000 process=cat proc_exepath=/usr/bin/cat parent
=sudo command=cat /etc/shadow terminal=34818 container_id=host container_name=ho
st container_image_repository= container_image_tag= k8s_pod_name=<NA> k8s_ns_nam
e=<NA>
Apr 25 22:24:38 mohammad-VMware-Virtual-Platform falco[28756]: 22:24:38.81105331
5: Warning Sensitive file opened for reading by non-trusted program | file=/etc/
shadow gparent=sudo gparent=bash gggparent=gnome-terminal- evt_type=openat user
=root user_uid=0 user_loginuid=1000 process=cat proc_exepath=/usr/bin/cat parent
=sudo command=cat /etc/shadow terminal=34820 container_id=host container_name=ho
st container_image_repository= container_image_tag= k8s_pod_name=<NA> k8s_ns_nam
e=<NA>
Apr 25 22:26:27 mohammad-VMware-Virtual-Platform falco[28756]: 22:26:27.12539832
3: Warning Sensitive file opened for reading by non-trusted program | file=/etc/
shadow gparent=sudo gparent=bash gggparent=gnome-terminal- evt_type=openat user
=root user_uid=0 user_loginuid=1000 process=cat proc_exepath=/usr/bin/cat parent
=sudo command=cat /etc/shadow terminal=34820 container_id=host container_name=ho
st container_image_repository= container_image_tag= k8s_pod_name=<NA> k8s_ns_nam
e=<NA>
systemd-oom:!*:20305:*****
kernoops:!*:20305:*****
whoopsie:!*:20305:*****
dnsmasq:!*:20305:*****
avahi:!*:20305:*****
tcpdump:!*:20305:*****
sssd:!*:20305:*****
speech-dispatcher:!*:20305:*****
cups-pk-helper:!*:20305:*****
fwupd-refresh:!*:20305:*****
saned:!*:20305:*****
geoclue:!*:20305:*****
cups-browsed:!*:20305:*****
hplip:!*:20305:*****
gnome-remote-desktop:!*:20305:*****
polkitd:!*:20305:*****
rtkit:!*:20305:*****
colord:!*:20305:*****
gnome-initial-setup:!*:20305:*****
gdm:!*:20305:*****
nm-openvpn:!*:20305:*****
mohammad:$6$WtKEAjTUqWbTqkUe$UuKFdR9Ld2cMQftadIkUs/WAuoMbm2KVe6vtnNaBj2RIa9r
UgXUA7k9iyd60t5jInbs99GKKSNoF5ez8gz770:20559:0:99999:7:::
mohammad@mohammad-VMware-Virtual-Platform: ~/Desktop$
```

8. Choosing, downloading, and enabling a cloud file storage service to offer it as a service from the cloud server to the client.

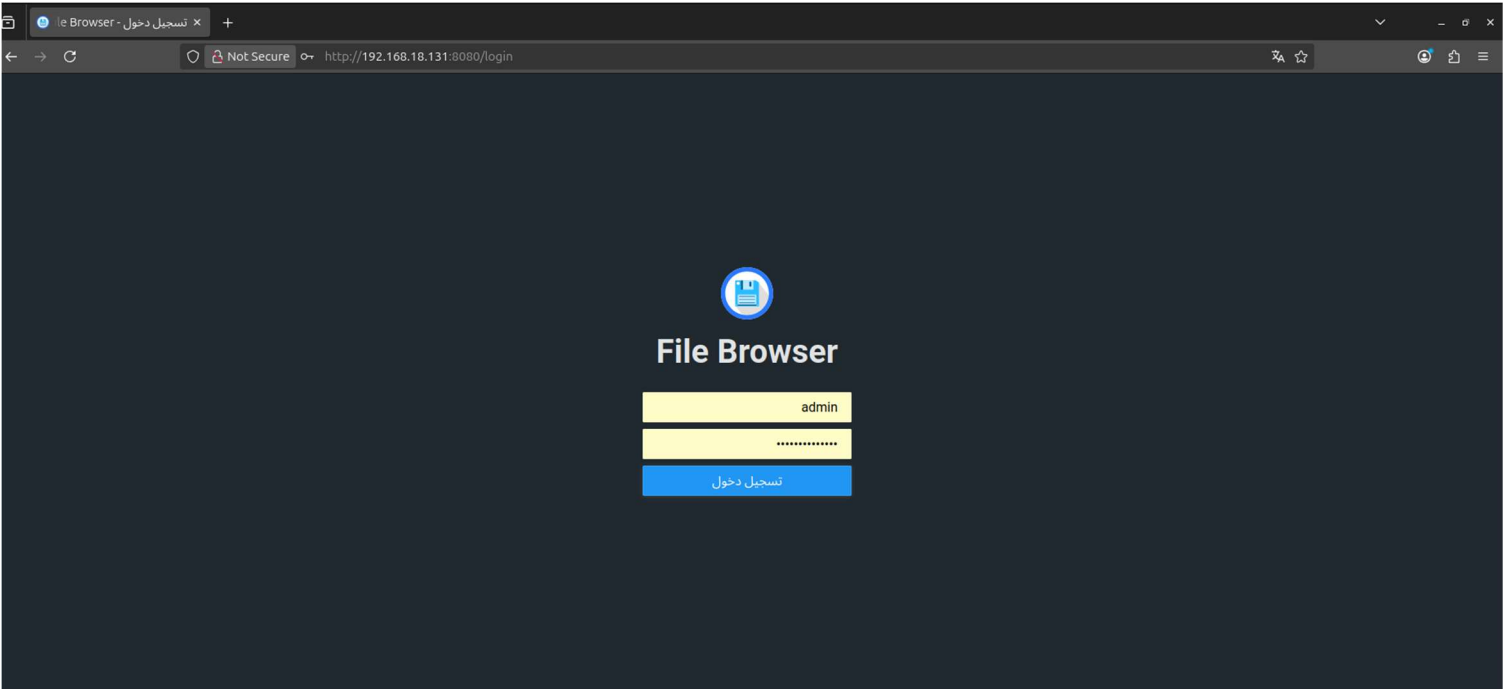
We downloaded from:

<https://github.com/filebrowser/filebrowser>

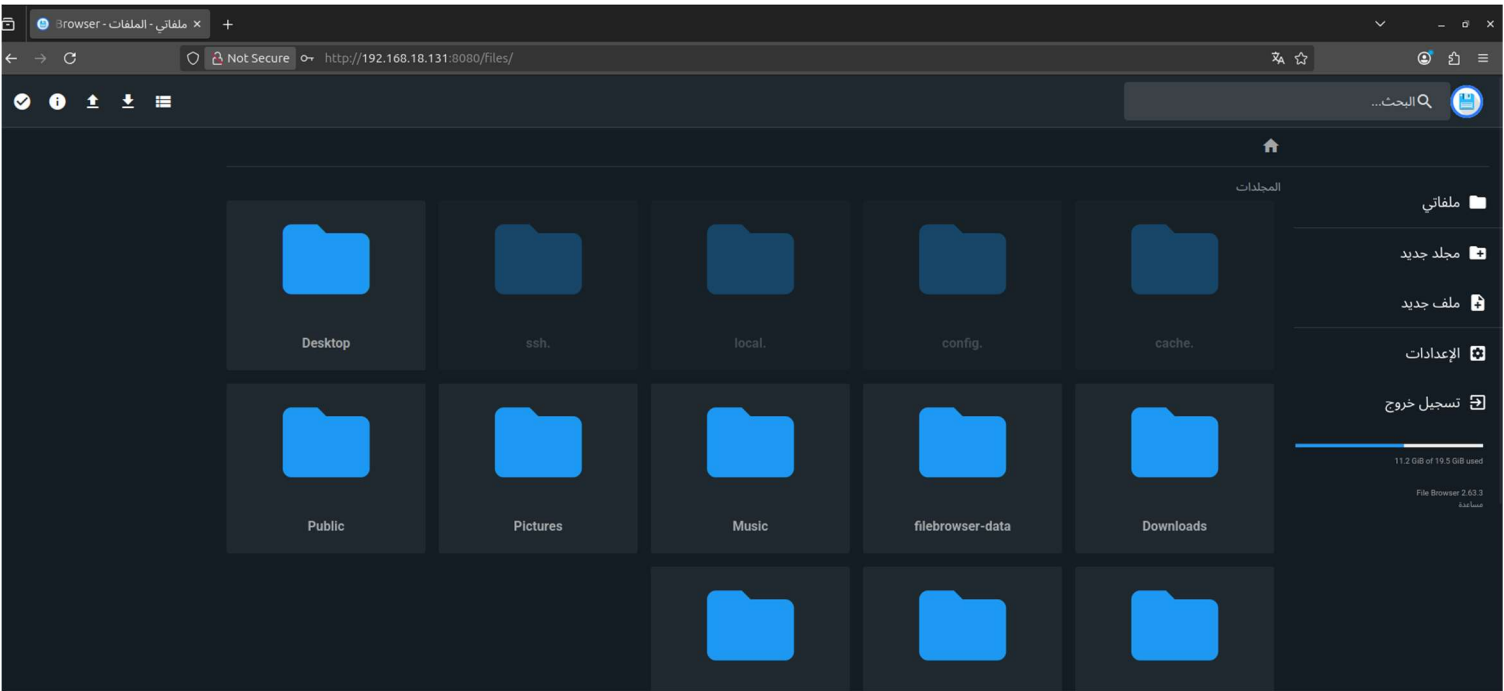
file browser status :

```
mohammad@mohammad-VMware-Virtual-Platform:~/Desktop$ sudo kubectl get pods -o wide
[sudo] password for mohammad:
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE
NOMINATED NODE   READINESS GATES
filebrowser-dffffdff5-dkn99        1/1     Running   1 (138m ago)  34h   10.42.1.7     worker1
<none>                             <none>
```

9.Choose username &password for file storage:



10.Download repo to manage file storage:



11. Download work measurement tools:

Stress.ng download :

Sudo apt-get update

sudo apt install stress-ng

K6 download:

sudo apt-get update

sudo apt-get install -y k6

K6 and stress-ng status :

```
mohammad@mohammad-VMware-Virtual-Platform:~/Desktop$ stress-ng --version
stress-ng, version 0.17.06 (gcc 13.2.0, x86_64 Linux 6.17.0-22-generic) 🖥️🔥
mohammad@mohammad-VMware-Virtual-Platform:~/Desktop$ k6 version
k6 v1.6.1 (commit/2ac2bb560e, go1.25.6, linux/amd64)
mohammad@mohammad-VMware-Virtual-Platform:~/Desktop$
```


12. Load testing using my tools k6 and stress-ng before carrying out an attack on the cloud server .

Stress-ng before attack:

```
mohammad@mohammad-VMware-Virtual-Platform: ~/Desktop

0[|||] 3.3%] Tasks: 139, 509 thr, 186 kthr; 1 running
1[||||] 8.1%] Load average: 0.15 0.26 0.37
Mem[|||||] 1.68G/7.71G Uptime: 00:20:03
Swp[|] 0K/3.78G]

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1649 root 20 0 1988M 587M 156M S 0.0 7.4 0:01.61 /usr/local/bin/k3s server
1687 root 20 0 1988M 587M 156M S 0.0 7.4 0:05.75 /usr/local/bin/k3s server
1688 root 20 0 1988M 587M 156M S 0.0 7.4 0:00.45 /usr/local/bin/k3s server
1689 root 20 0 1988M 587M 156M S 0.0 7.4 0:00.94 /usr/local/bin/k3s server
1690 root 20 0 1988M 587M 156M S 0.0 7.4 0:00.00 /usr/local/bin/k3s server
1719 root 20 0 1988M 587M 156M S 0.0 7.4 0:00.00 /usr/local/bin/k3s server
1737 root 20 0 1988M 587M 156M S 2.7 7.4 0:19.77 /usr/local/bin/k3s server
1785 root 20 0 1988M 587M 156M S 0.0 7.4 0:12.37 /usr/local/bin/k3s server
2019 root 20 0 1988M 587M 156M S 0.0 7.4 0:11.30 /usr/local/bin/k3s server
2042 root 20 0 1988M 587M 156M S 4.0 7.4 0:20.93 /usr/local/bin/k3s server
2052 root 20 0 1988M 587M 156M S 0.0 7.4 0:00.42 /usr/local/bin/k3s server
2053 root 20 0 1988M 587M 156M S 0.0 7.4 0:01.04 /usr/local/bin/k3s server
2054 root 20 0 1988M 587M 156M S 0.0 7.4 0:22.38 /usr/local/bin/k3s server
2055 root 20 0 1988M 587M 156M S 0.0 7.4 0:02.17 /usr/local/bin/k3s server
2056 root 20 0 1988M 587M 156M S 0.0 7.4 0:02.02 /usr/local/bin/k3s server
3683 root 20 0 1988M 587M 156M S 2.0 7.4 0:15.62 /usr/local/bin/k3s server
3684 root 20 0 1988M 587M 156M S 0.0 7.4 0:00.04 /usr/local/bin/k3s server
4250 mohammad 20 0 3736M 266M 130M S 0.0 3.4 0:15.59 /usr/bin/gnome-shell
4264 mohammad 20 0 3736M 266M 130M S 0.0 3.4 0:00.00 /usr/bin/gnome-shell
4265 mohammad 20 0 3736M 266M 130M S 0.0 3.4 0:00.02 /usr/bin/gnome-shell
4267 mohammad 20 0 3736M 266M 130M S 0.0 3.4 0:00.50 /usr/bin/gnome-shell
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit
```

K6 before attack :

```
mohammad@mohammad-VMware-Virtual-Platform: ~/Desktop

output: -

scenarios: (100.00%) 1 scenario, 50 max VUs, 1m0s max duration (incl. graceful stop):
* default: 50 looping VUs for 30s (gracefulStop: 30s)

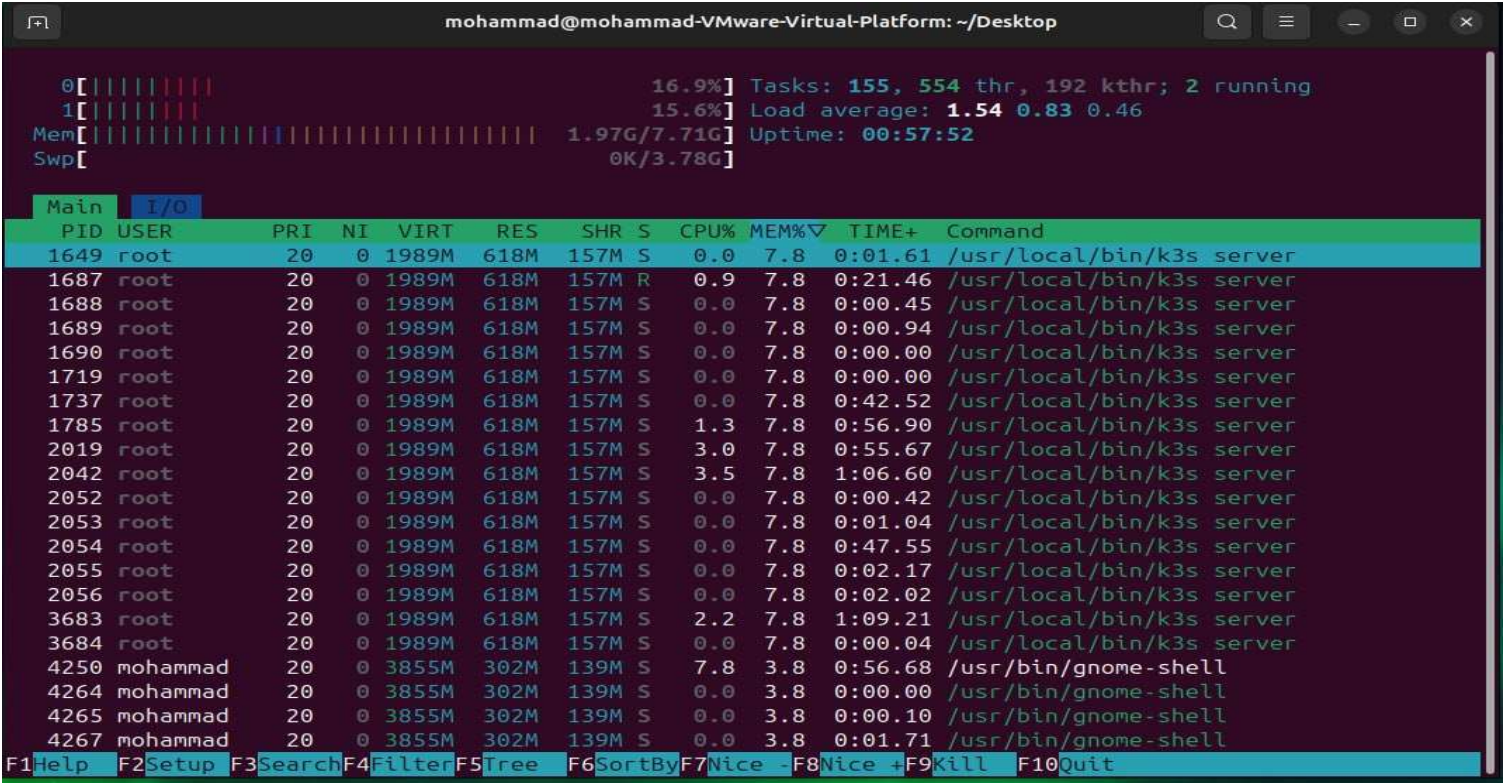
TOTAL RESULTS

HTTP
http_req_duration..... avg=4.78ms min=702.43µs med=2.79ms max=34.92ms p(90)=11.23ms p(95)=15.24ms
{ expected_response:true }... avg=4.78ms min=702.43µs med=2.79ms max=34.92ms p(90)=11.23ms p(95)=15.24ms
http_req_failed..... 0.00% 0 out of 1500
http_reqs..... 1500 49.559961/s

EXECUTION
iteration_duration..... avg=1s min=1s med=1s max=1.04s p(90)=1.01s p(95)=1.02s
iterations..... 1500 49.559961/s
vus..... 50 min=50 max=50
vus_max..... 50 min=50 max=50

NETWORK
data_received..... 8.8 MB 292 kB/s
data_sent..... 114 kB 3.8 kB/s
```

Stress-ng after attack:



K6 after attack :

